Lecture 29 — Strict Cardinality and Pigeonhole

CS 173: Discrete Structures · $|A| < |B|$, $|A| > |B|$

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1 Strict Cardinality

Definition 1.1 ($|A| < |B|$)**.** For any sets A, B:

$$|A| < |B| \iff \exists f\,(f : A \to B \wedge f \text{ injective}) \wedge \forall g\,(g : A \to B \Rightarrow g \text{ not surjective}).$$

Definition 1.2 ($|A| > |B|$)**.** For any sets A, B:

$$|A| > |B| \iff \exists g\,(g : A \to B \wedge g \text{ surjective}) \wedge \forall f\,(f : A \to B \Rightarrow f \text{ not injective}).$$

Remark. Each definition pairs an existential witness with a universal failure: A fits inside B but cannot cover it, or covers B but cannot embed in it. The universal clause is essential — a single non-surjective function (e.g. a constant) proves nothing on its own.

2 Pigeonhole Theorem

Theorem 2.1 (Pigeonhole)**.** For any sets A, B:

(i) $|A| > |B| \Rightarrow \forall f\,(f : A \to B \Rightarrow f \text{ not injective})$.

(ii) $|A| < |B| \Rightarrow \forall f\,(f : A \to B \Rightarrow f \text{ not surjective})$.

Remark. Pigeonhole upgrades the universal clause of each definition into a theorem you can apply once strict inequality is established.

3 Quantitative Pigeonhole

The basic theorem says some fibre is non-singleton. The quantitative form pins down *how* crowded that fibre must be.

Theorem 3.1 (Quantitative Pigeonhole)**.** Let $|A| = n \in \mathbb{N}^+$ and $|B| = k \in \mathbb{N}^+$. Then for any function $f : A \to B$:

$$\exists b \in B \left(|\{\, a \in A \mid f(a) = b \,\}| \ \geq\ \left\lceil \tfrac{n}{k} \right\rceil \right).$$

Remark. When $k \nmid n$, the bound rewrites as $\lceil \frac{n}{k} \rceil = \lfloor \frac{n}{k} \rfloor + 1$ — the form written on the board. When $k \mid n$, ceiling and floor agree and equal n/k.

Remark (Sanity check)**.** If every fibre had size $< \lceil n/k \rceil$, i.e. at most $\lceil n/k \rceil - 1$, the total $\sum_{b\in B} |f^{-1}(\{b\})|$ would be at most $k(\lceil n/k \rceil - 1) < n$ — but this sum equals $|A| = n$. Contradiction.

4 Finite Sets: One Witness Suffices

Proposition 4.1 (Finite shortcut)**.** Let A, B be finite. Then:

(i) If $f : A \to B$ is injective and not surjective, then $|A| < |B|$.

(ii) If $g : A \to B$ is surjective and not injective, then $|A| > |B|$.

Proof. **(i)** f witnesses the existential clause. For the universal: if some $g : A \to B$ were surjective, then $|A| \leq |B|$ (from f) and $|A| \geq |B|$ (from g) give $|A| = |B|$, forcing f to be a bijection — contradicting non-surjectivity.

(ii) Symmetric: a rival injection f would force g to be a bijection, contradicting non-injectivity. □

Caution (Infinite sets break the shortcut)**.** Take $A = B = \mathbb{N}$, $f(n) = 2n$: injective, not surjective, yet $|\mathbb{N}| = |\mathbb{N}|$. The identity $g(n) = n$ kills the universal clause. For $|\mathbb{N}| < |\mathbb{R}|$ and similar infinite strict comparisons, the universal clause must be proved directly — the job of Cantor's diagonal.

5 Summary

1. Strict cardinality = existential witness $\wedge$ universal failure of the opposite kind of function.
2. Pigeonhole supplies the universal clause once strict inequality holds.
3. Quantitative form: with $|A| = n$, $|B| = k$, some fibre has size at least $\lceil n/k \rceil$.
4. Finite case: a non-surjective injection witnesses $|A| < |B|$; a non-injective surjection witnesses $|A| > |B|$.
5. Infinite case: the universal clause needs a direct proof.